Bearing-only maritime localization



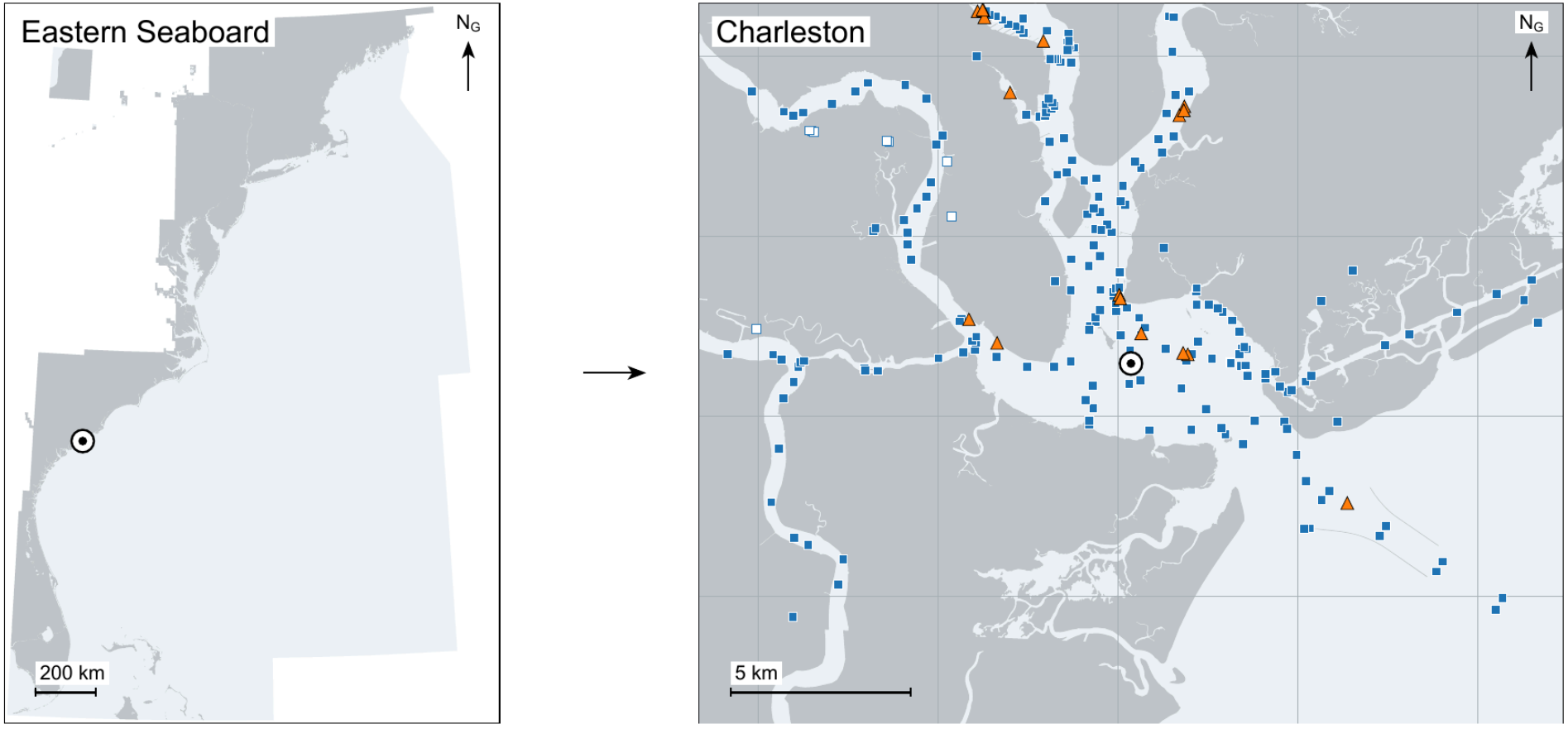


Charleston · B17

Chart aid AIS Land Water Unknown

J.C. Vaught

Where is the observer?



Range uncertainty



Calculation on one actual mapped-object ray

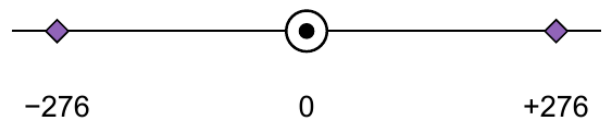
±10%

Range sensitivity

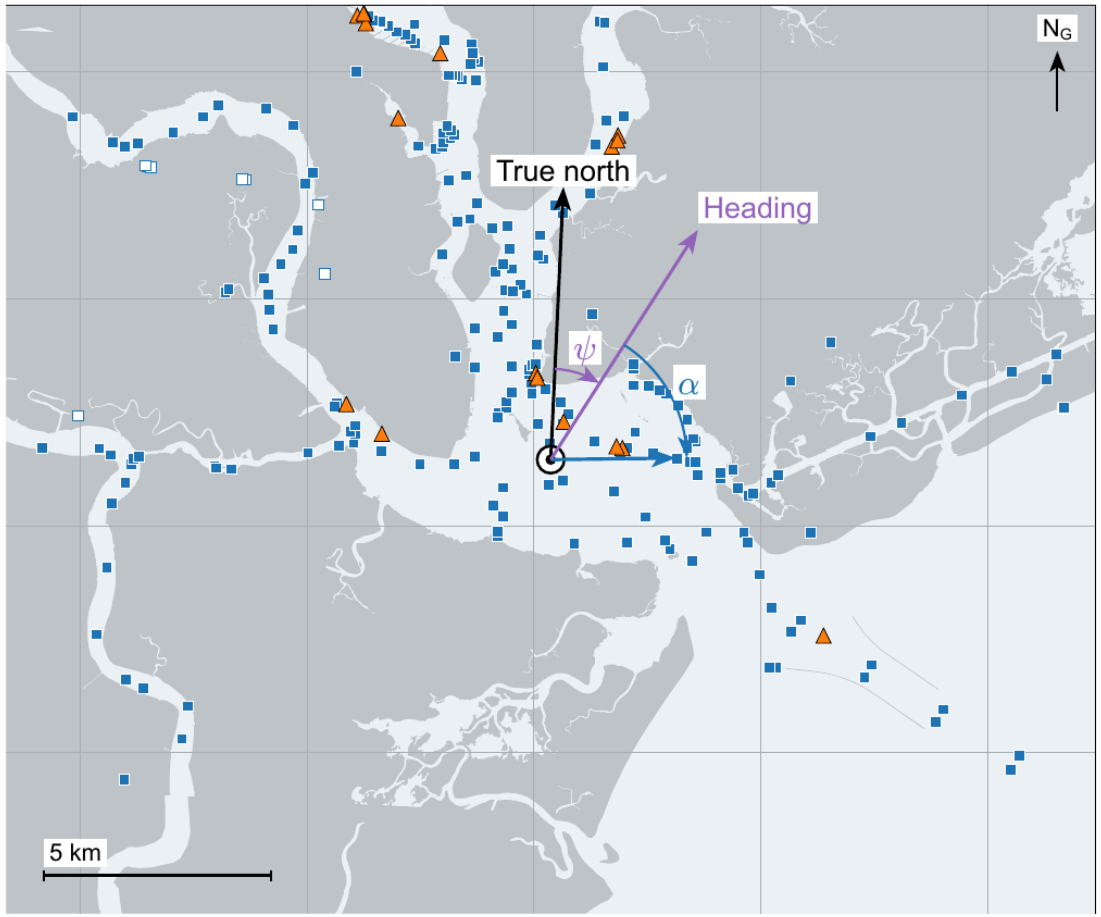
±276 m

Calculated displacement

Along the observed ray · meters



Bearings relative to true north



$$\beta = \mathrm{wrap}(\psi + \alpha)$$

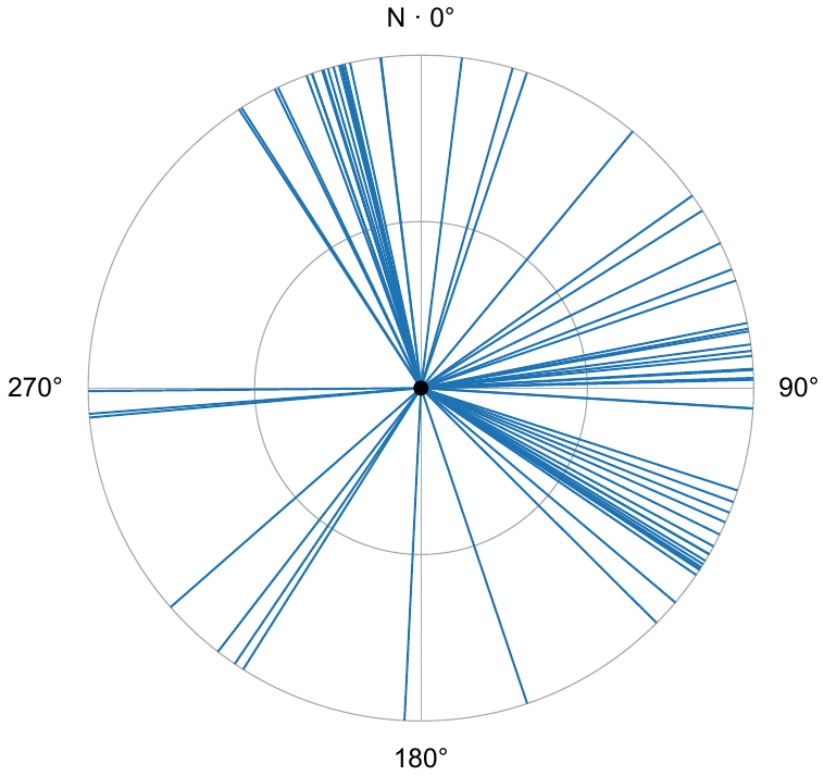
ψ · known heading

α · relative bearing

β · global bearing

Heading-conversion example · calculated $\psi = 30°$

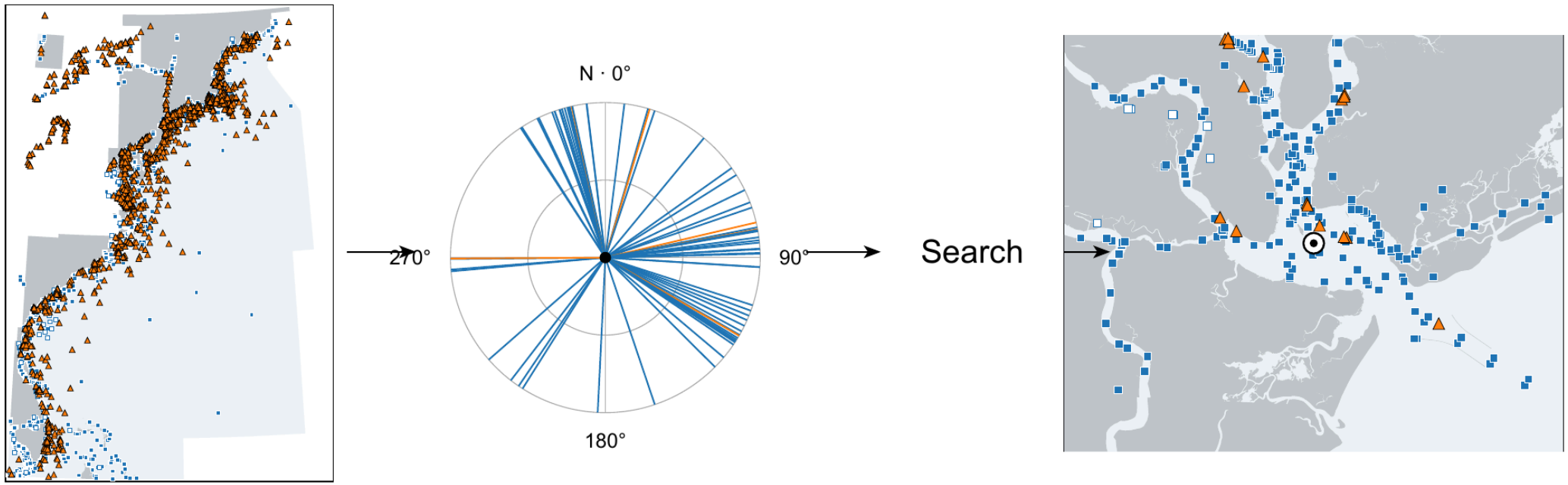
Bearings without supplied object identities



64 chart-aid bearings



The system



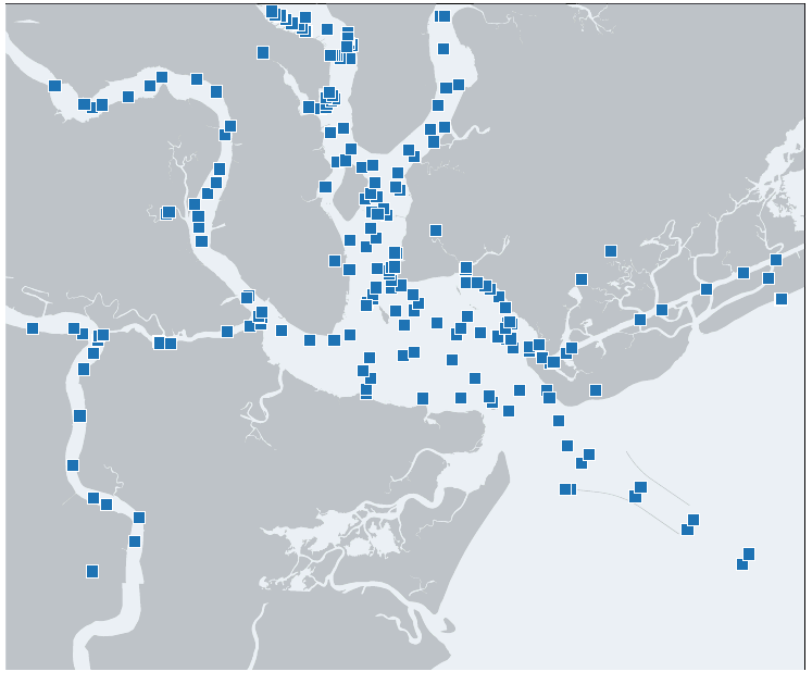
Catalog

Observations

Location

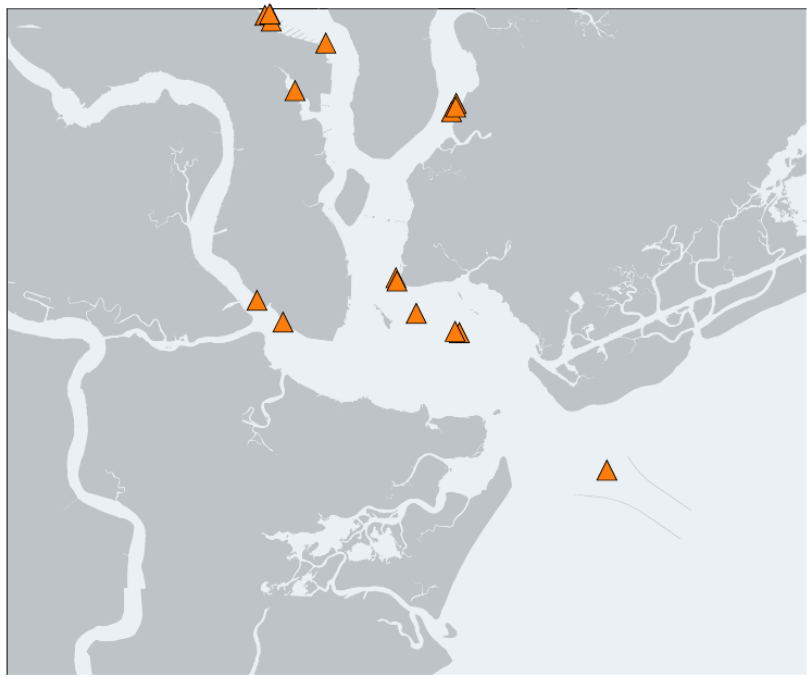
Current localization objective · chart-aid bearings

Geographic data sources



NOAA ENC

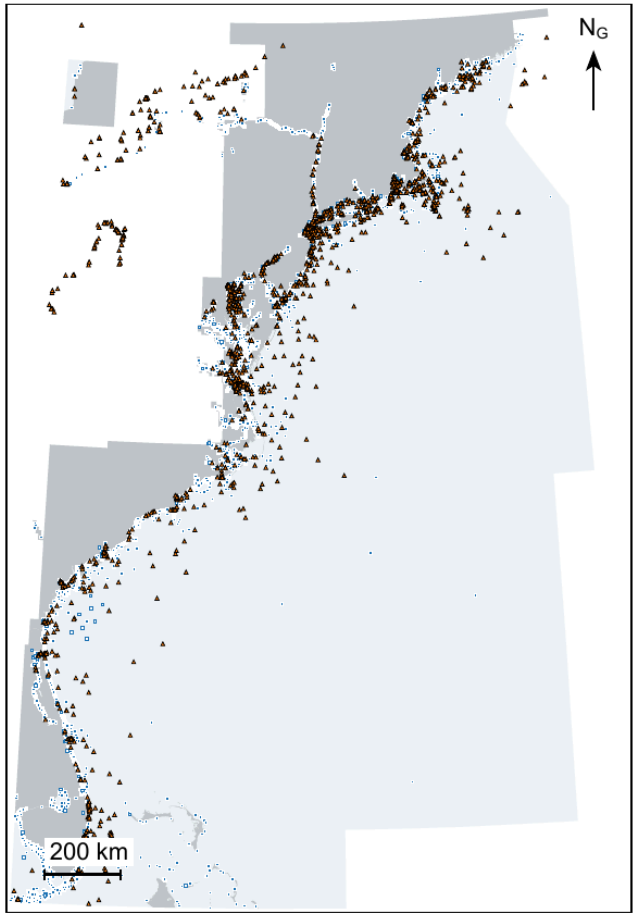
Chart features and shoreline



Archived AIS

NOAA vessel-position snapshot

The Eastern Seaboard catalog



32,168

Canonical chart locations

2,268

AIS-source records

Charleston’s mapped objects



Chart aid AIS Land Water Unknown

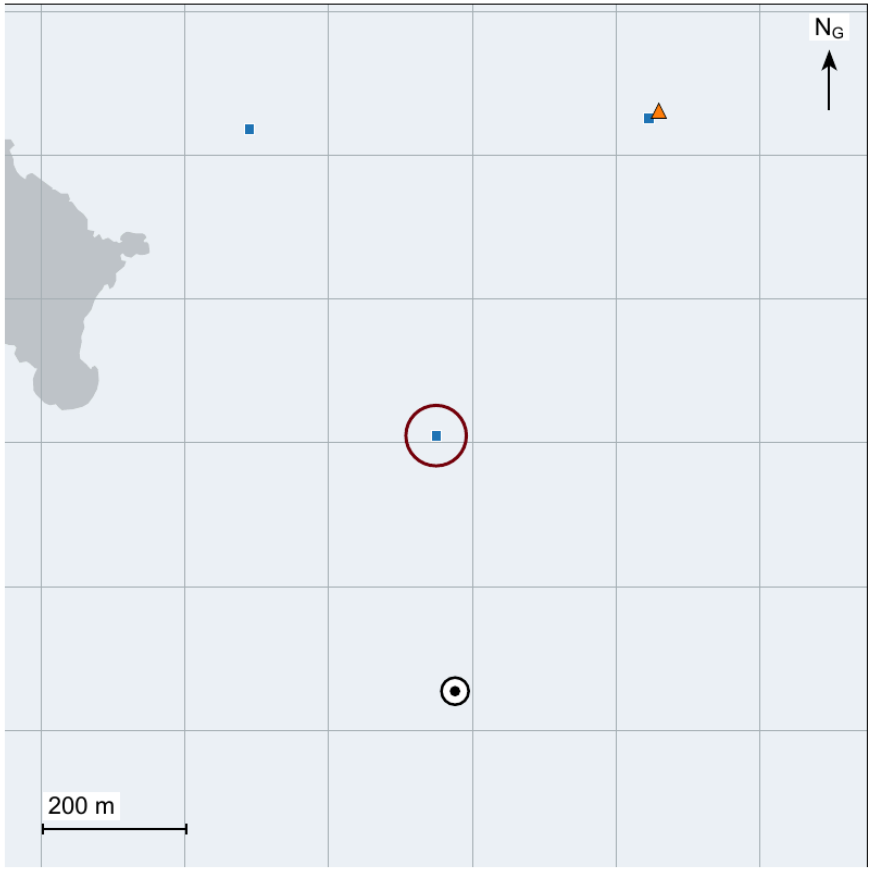
220

Chart locations

17

AIS records

Three components, one location



3 → 1

Co-located source records

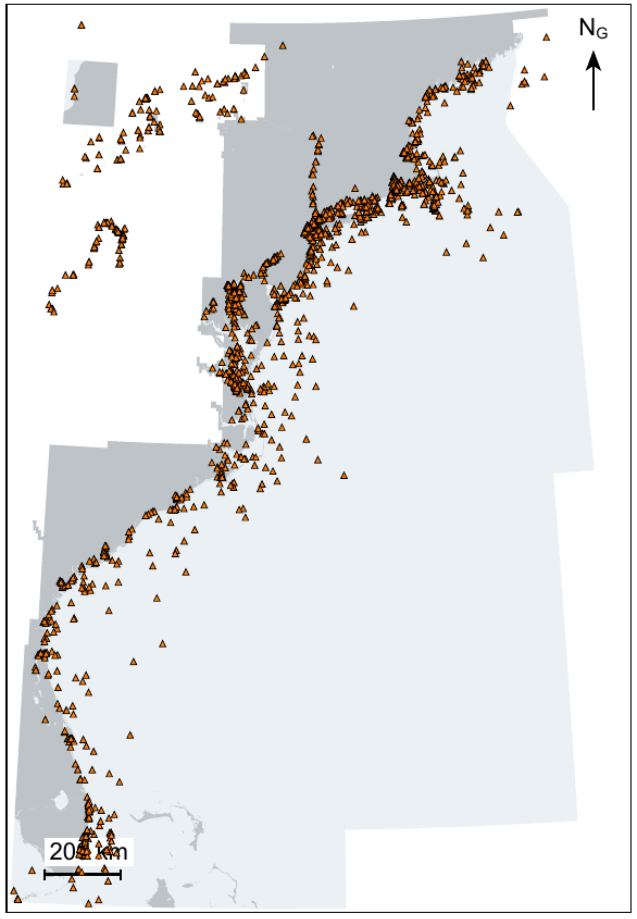
Beacon · light · daymark

32.7731847°, −79.9054558°

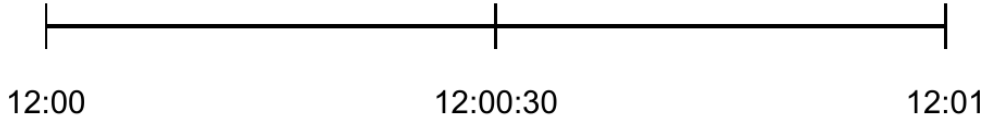
64,400 → 32,168

Raw rows → canonical chart locations

The AIS snapshot



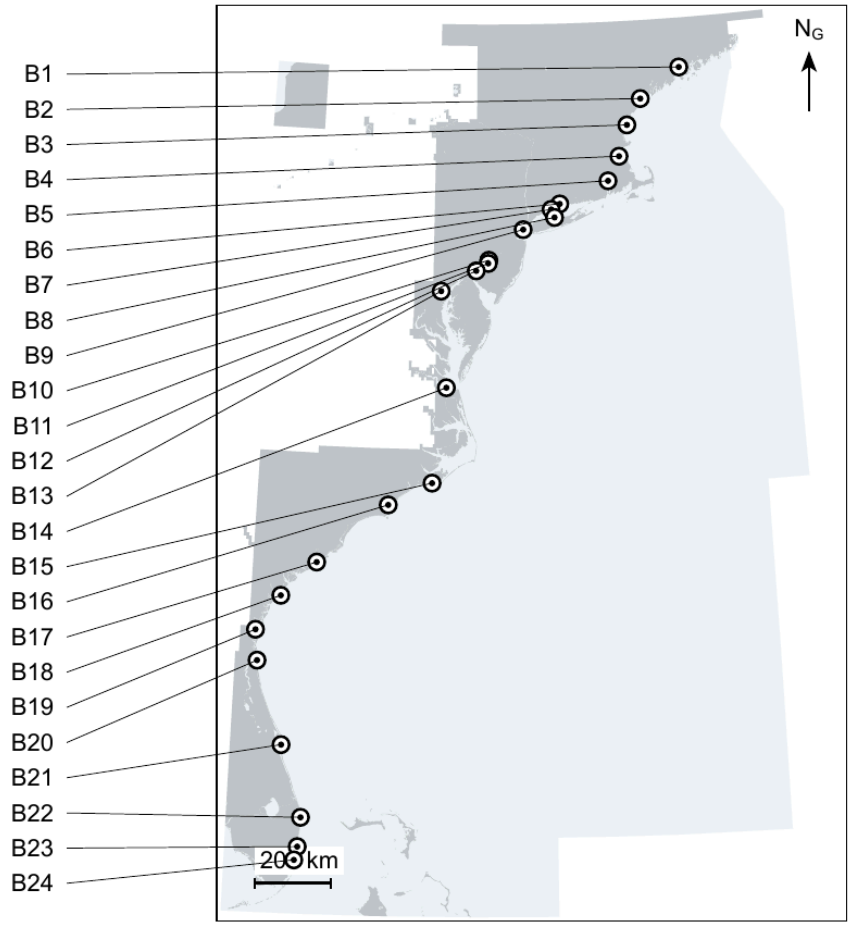
15 July 2023



2,268 retained records

Nearest report to midpoint per identity · UTC

Test locations



24

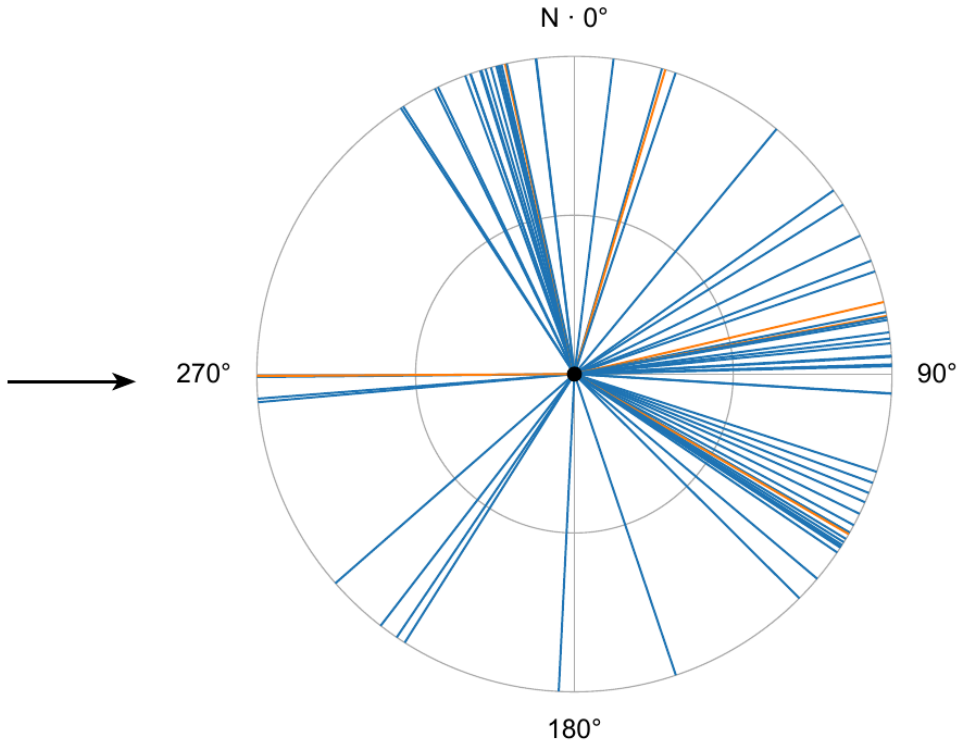
Test locations

Charleston · B17

Completed coast-wide search example

What the simulator produces





64 aids + 6 vessel reports

Chart aid · AIS · Land · Water · Unknown

VIDEO PLACEHOLDER · saved scene → bearing observations

Sensor configuration



360°

Field of view

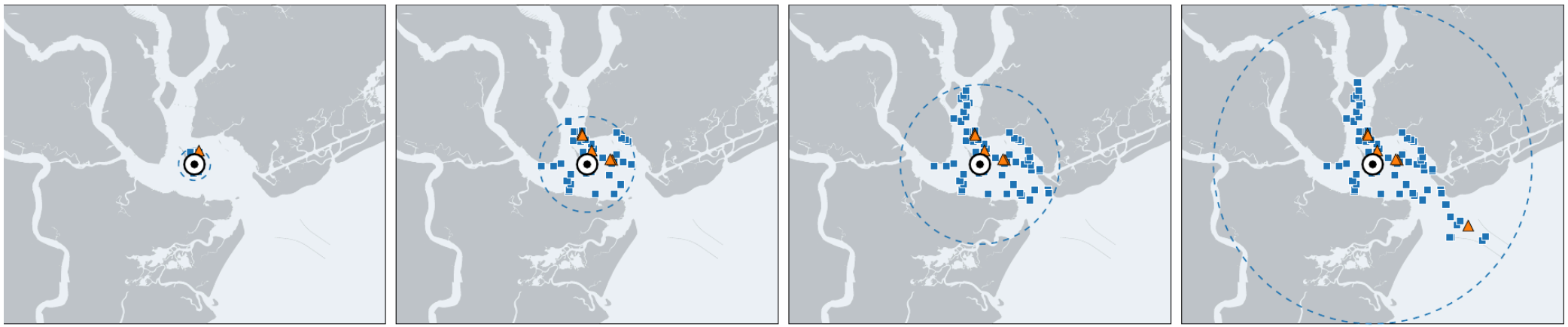
10 km

Visibility radius

0.1° grouping

±0.5° error bound

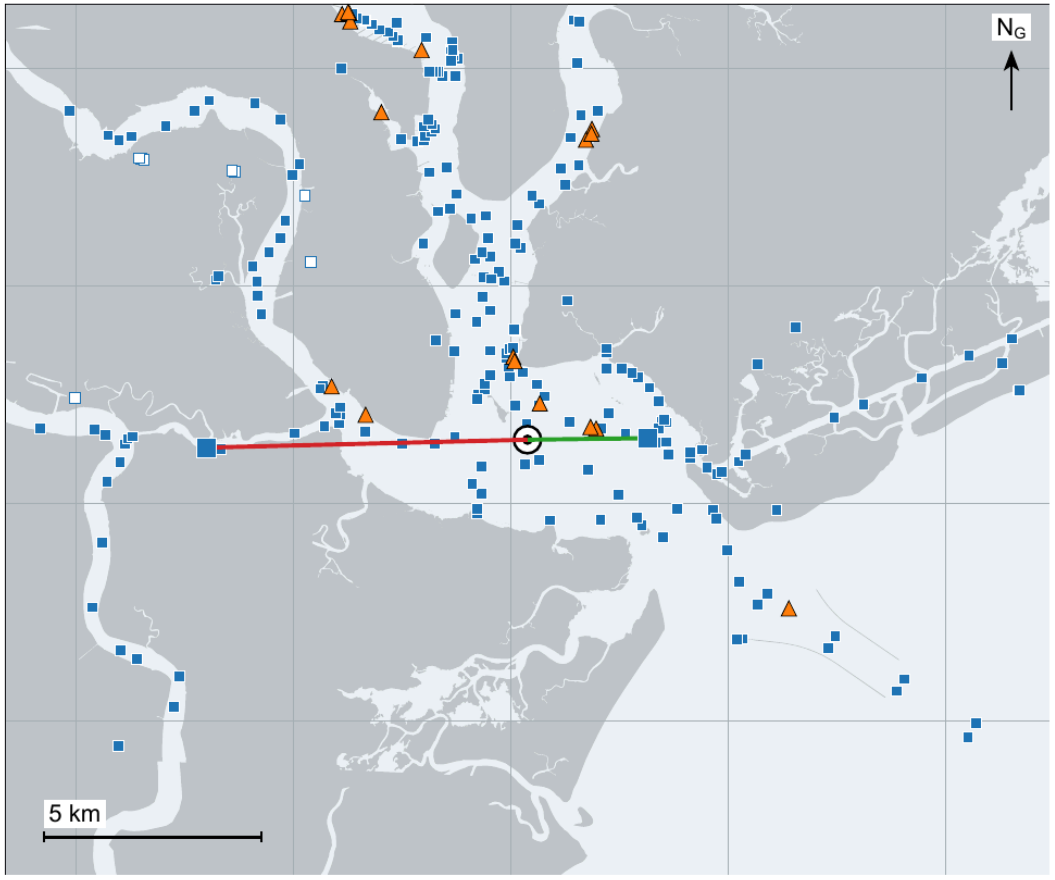
Visibility range changes the scene



1 km	3 km	5 km	10 km
6 / 6	45 / 54	70 / 102	80 / 200

Visible / in-range physical objects · saved line-of-sight classifications

Land blocks sight lines



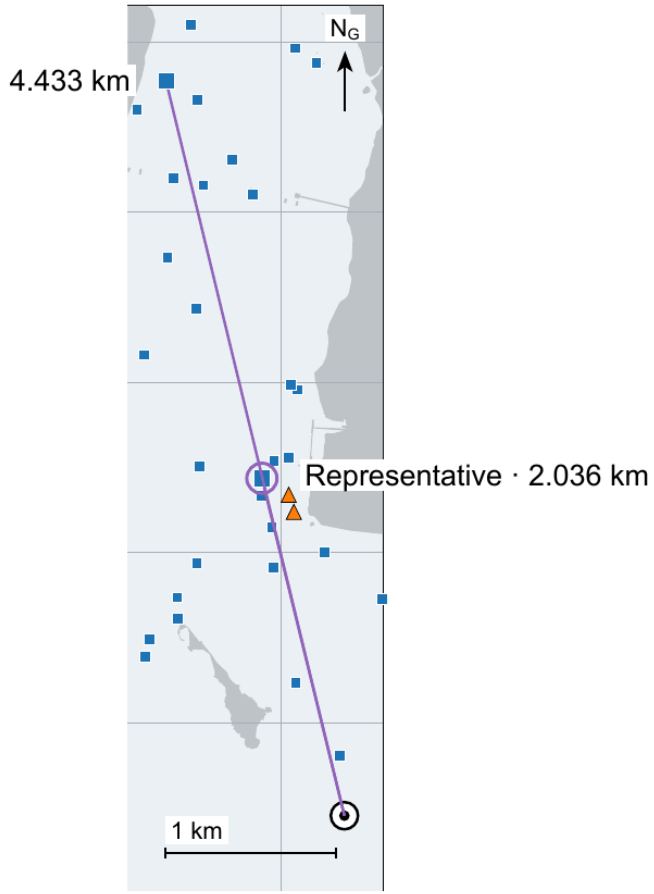
Visible

2.750 km

Land blocked

7.372 km

Angular merging



80 → 79

Visible objects → angular groups

Bearing · magnified

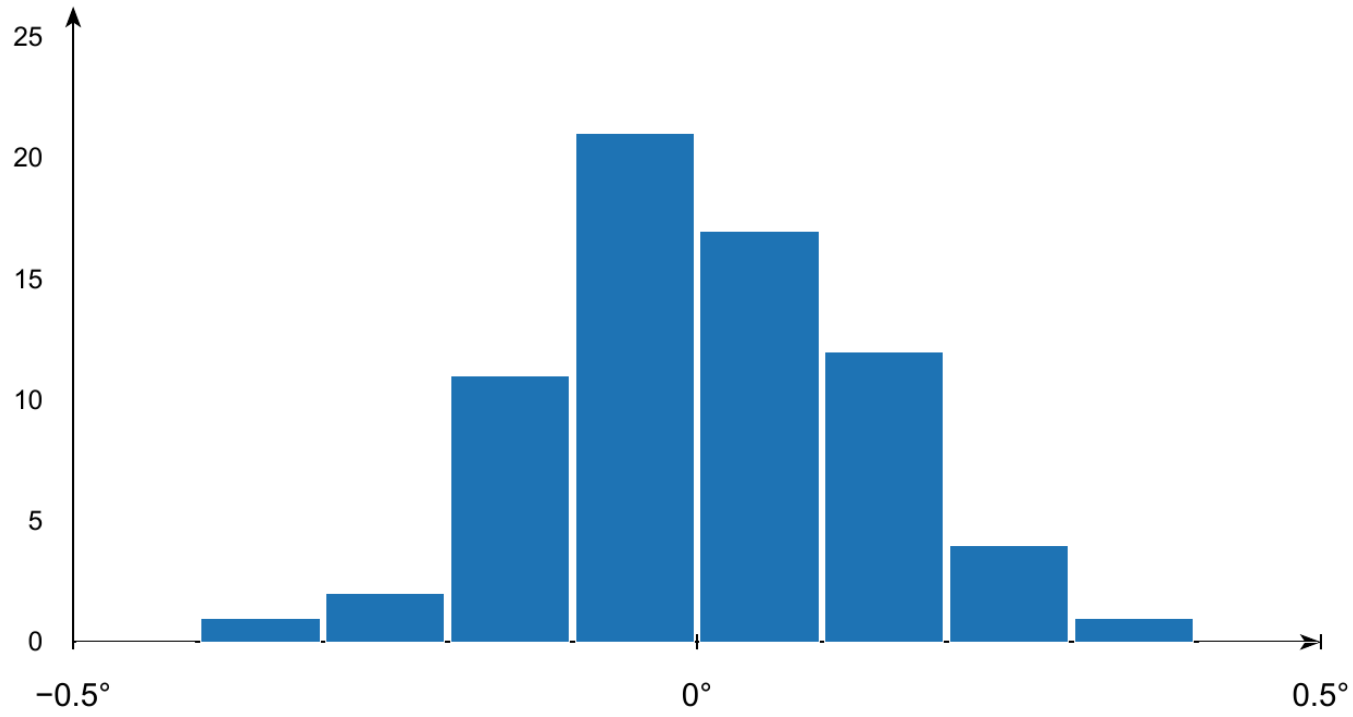
343.594°

343.685°

$\Delta\beta = 0.09147°$

0.1° grouping threshold

Bounded angular noise



Recovered errors · 69 simulated physical-object reports

1/6°

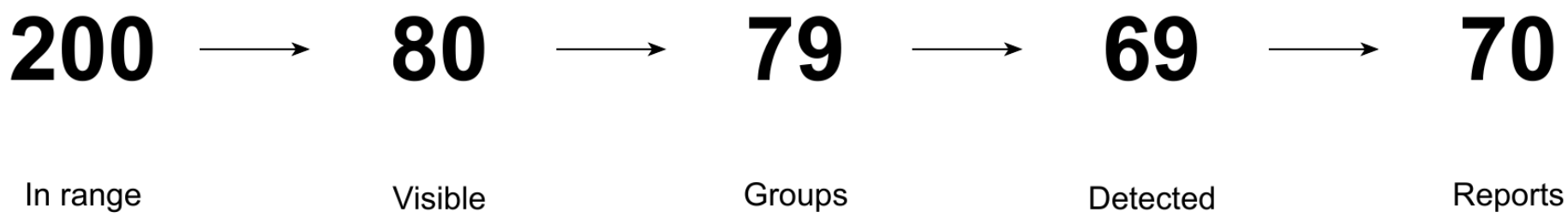
Underlying standard deviation

±0.5°

Hard bearing-error bound

Missed objects and extra vessel reports

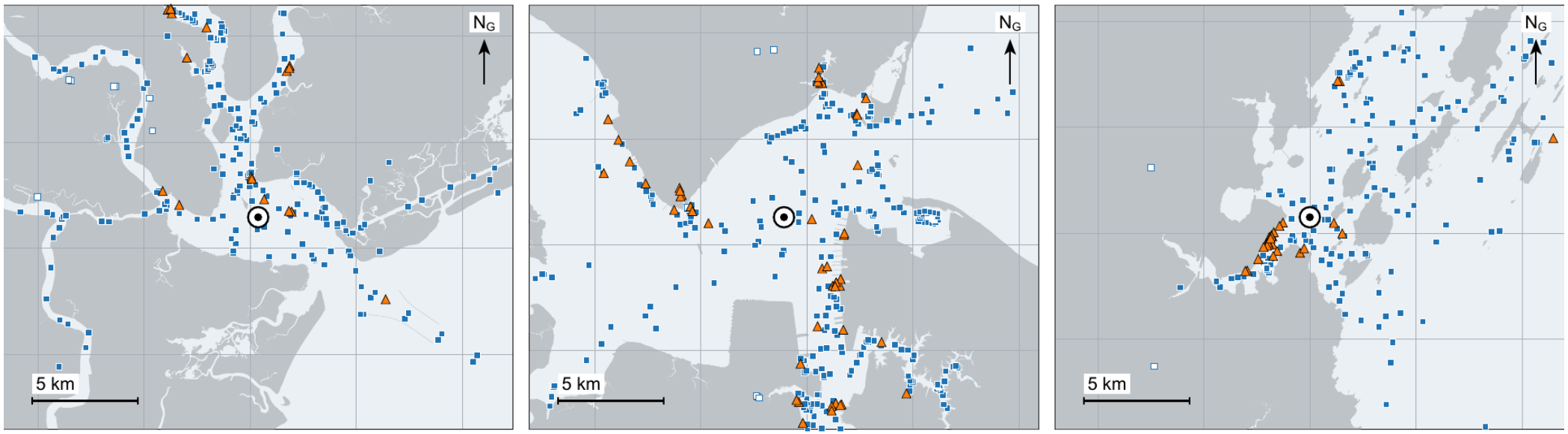
1. 1 extra vessel report



64 aid reports + 6 vessel reports

10 missed groups · no extra chart-aid reports

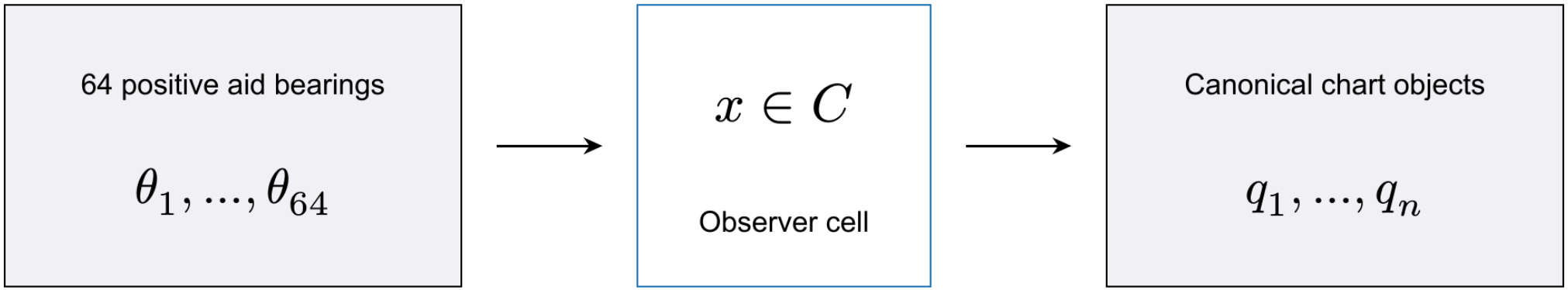
Worked harbor examples



Charleston | Norfolk | Portland

Real map/catalog crops · only Charleston has the completed localization result shown here.

What the search must establish



$$\beta_j(x) = \mathrm{azi}_{\mathrm{WGS84}}(x, q_j)$$

Global bearings. Object identities are not supplied.

Exhaustive search, even across multiple GPUs

$$W = 1644.676 \text{ km}$$

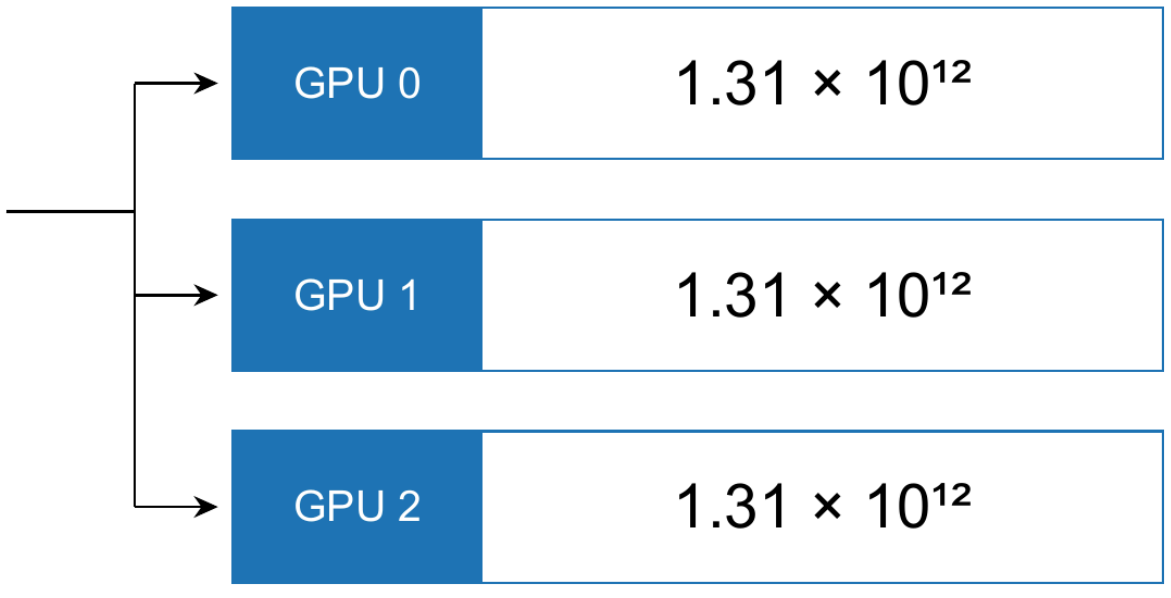
$$H = 2388.459 \text{ km}$$

$$N = \left\lceil \frac{W}{1\text{ m}} \right\rceil \left\lceil \frac{H}{1\text{ m}} \right\rceil$$

$$3.93 \times 10^{12}$$

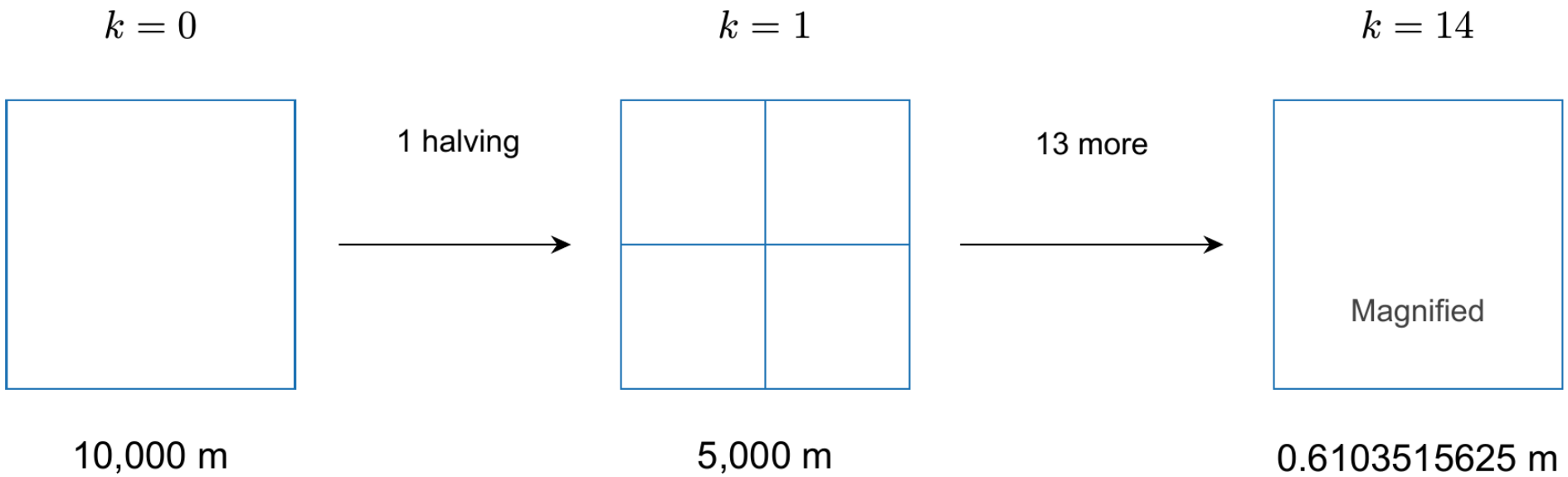
1 m grid positions

Ideal equal division



Calculated counts only. No exhaustive-search runtime was measured.

Coarse-to-fine search



$$h_k = \frac{10000}{2^k} \text{ m}$$

Mathematical subdivision. Every retained nonterminal cell is split into four.

Pruning

Every valid explanation in the cell must remain a possible edge.

Certified failure rejects. Unsupported bounds retain.

Whole cell is known land

$$C \subseteq L_{\text{known}}$$

One report has no object

$$\exists i : \deg_C(i) = 0$$

Too few distinct possibilities

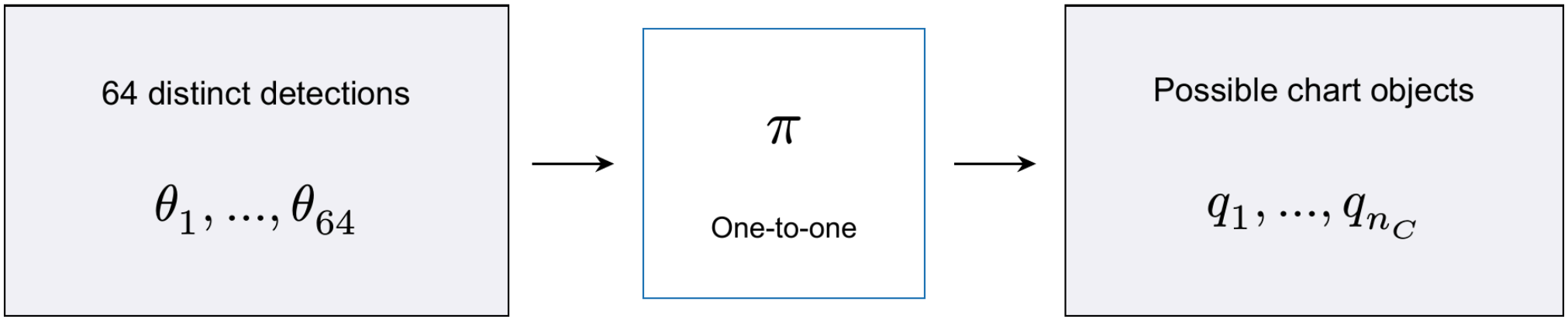
$$n_C < 64$$

No simultaneous injection

$$\nu(E_C) < 64$$

Whole-cell tests use positive reports. Unseen objects are not penalties.

Distinct markers must explain distinct reports



$$i \neq k \Rightarrow \pi(i) \neq \pi(k)$$

$$|N_C(S)| < |S| \Rightarrow \nu(E_C) < 64$$

Equations only. Actual rejection adjacency awaits a recorded witness.

Deterministic marker scoring

$$r_{ij}(x) = \left|\mathrm{wrap}_{180}\left(\theta_i - \beta_j(x)\right)\right|$$

64 injective exact matches

$$r_{i\pi(i)}(x) < \varepsilon_i - \delta_{\mathrm{num}}$$

Known water and selected visibility

Flags = 0

Select by summed residual

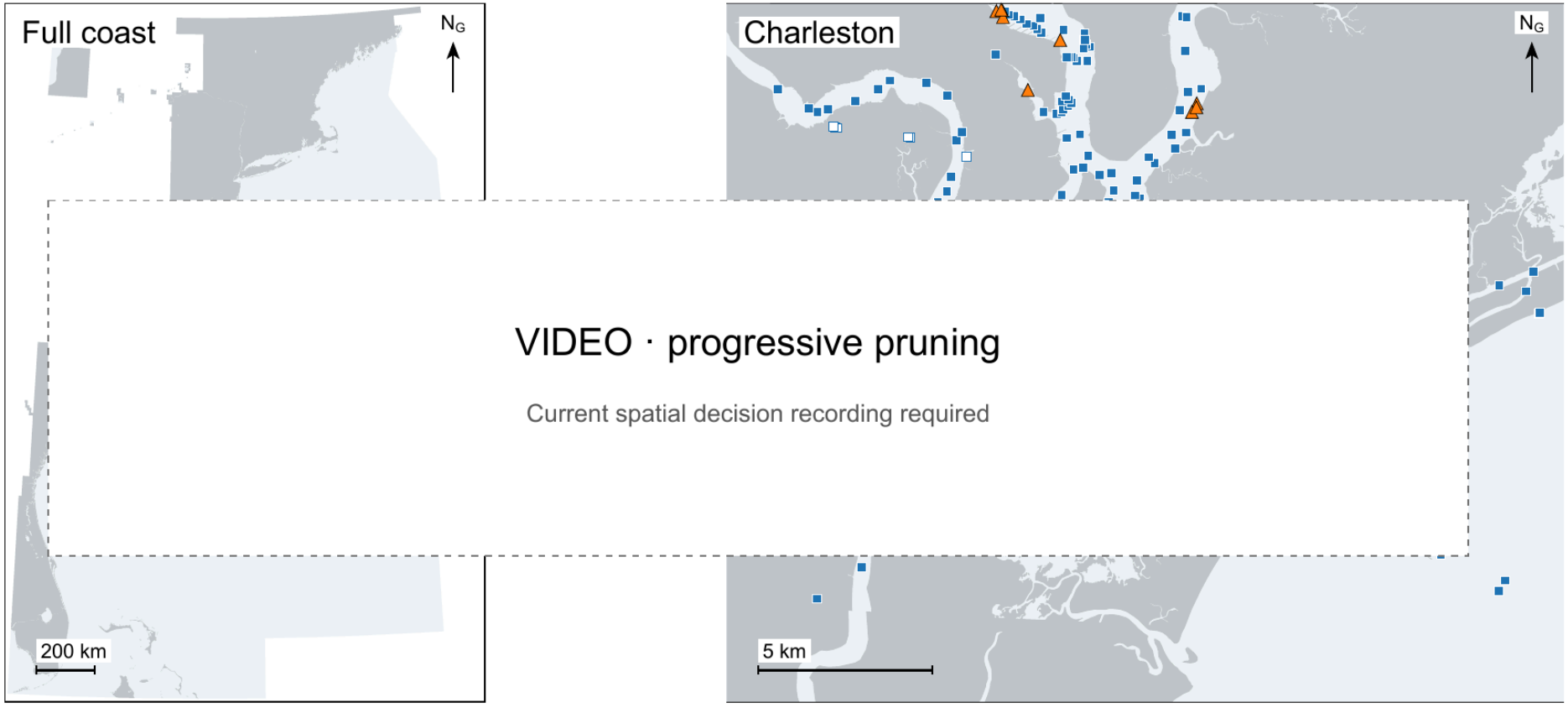
$$J(x) = \sum_{i=1}^{64} r_{i\pi(i)}(x)$$

$$\hat{x} = \mathrm{argmin}_{x \in X_{\mathrm{consistent}}} J(x)$$

$$J(\hat{x}) = 6.2809796\ ^\circ$$

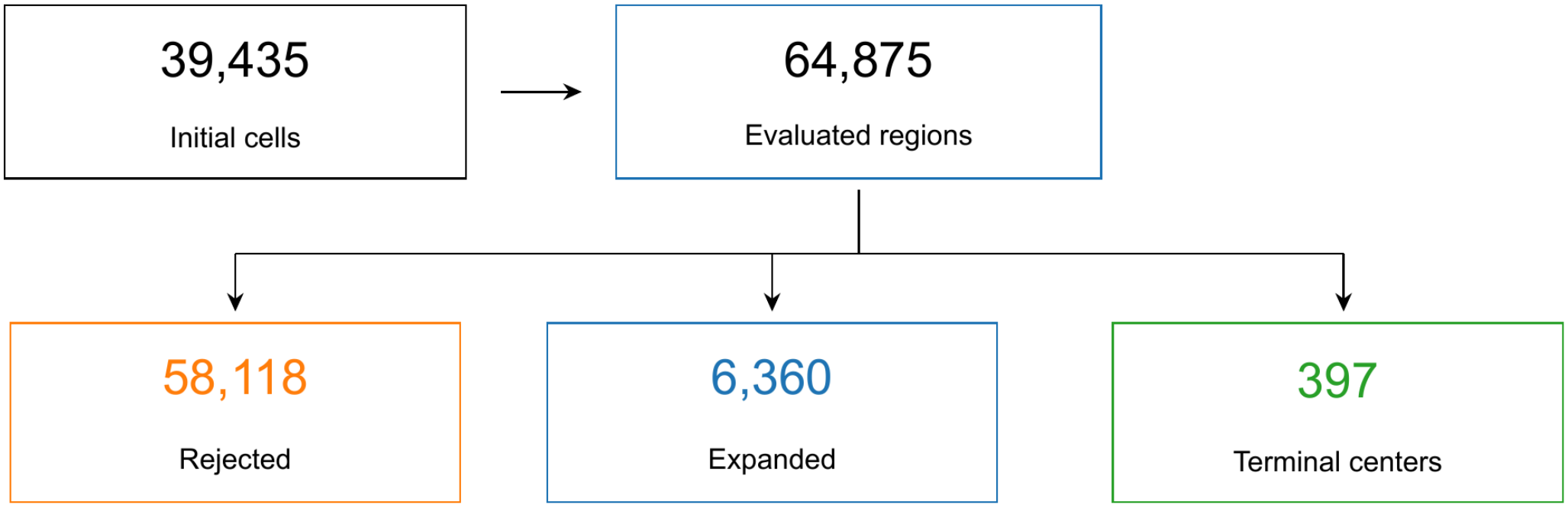
Current frozen estimate. Total residual over 64 positive aid reports.

The coast-wide search contracts toward Charleston



VIDEO PLACEHOLDER · synchronized pruning trace

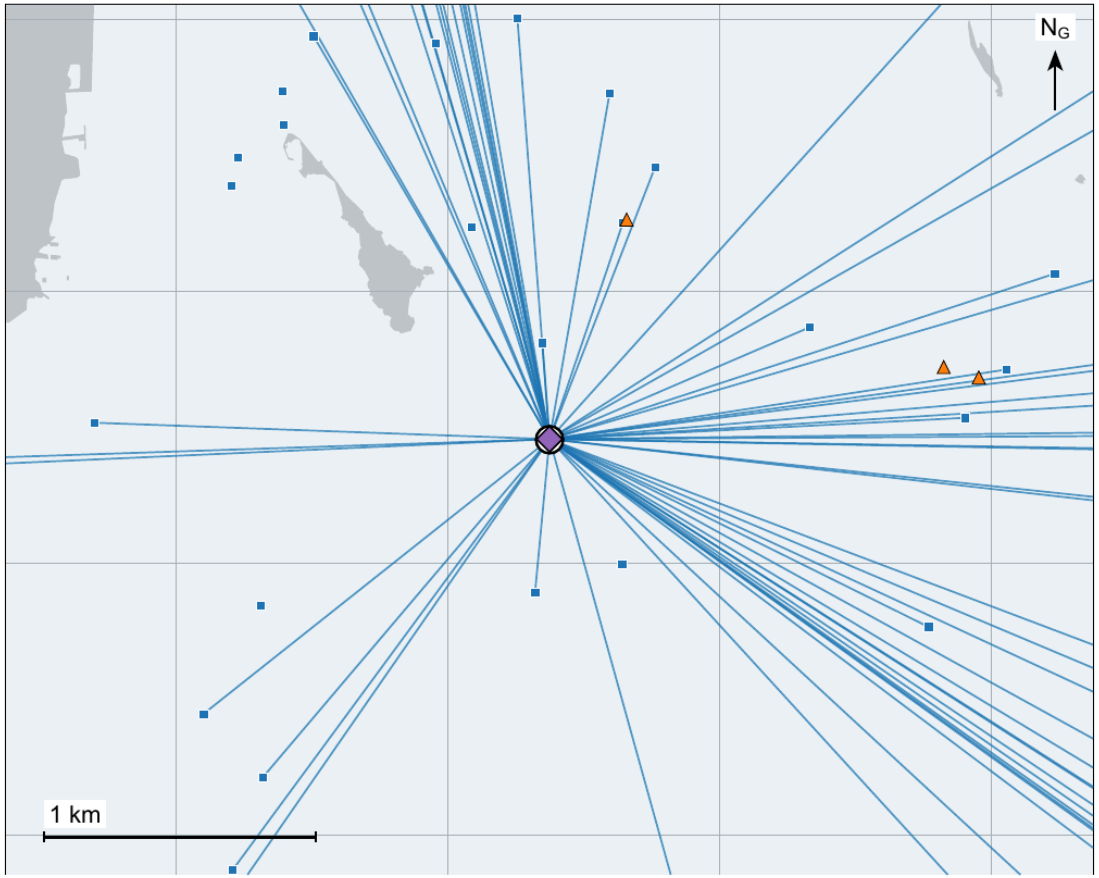
Recorded search workload



$$58118 + 6360 + 397 = 64875$$

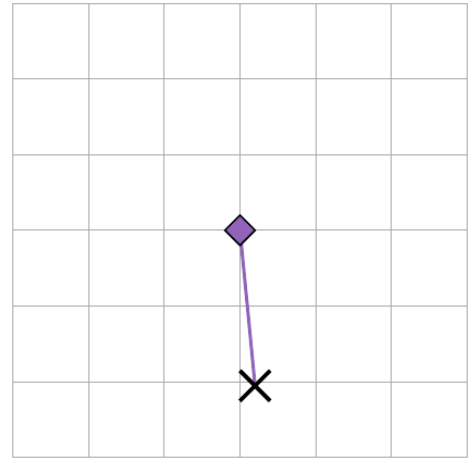
$$39435 + 4 \times 6360 = 64875$$

The selected Charleston position



Matched rays clipped to harbor crop

1 m grid



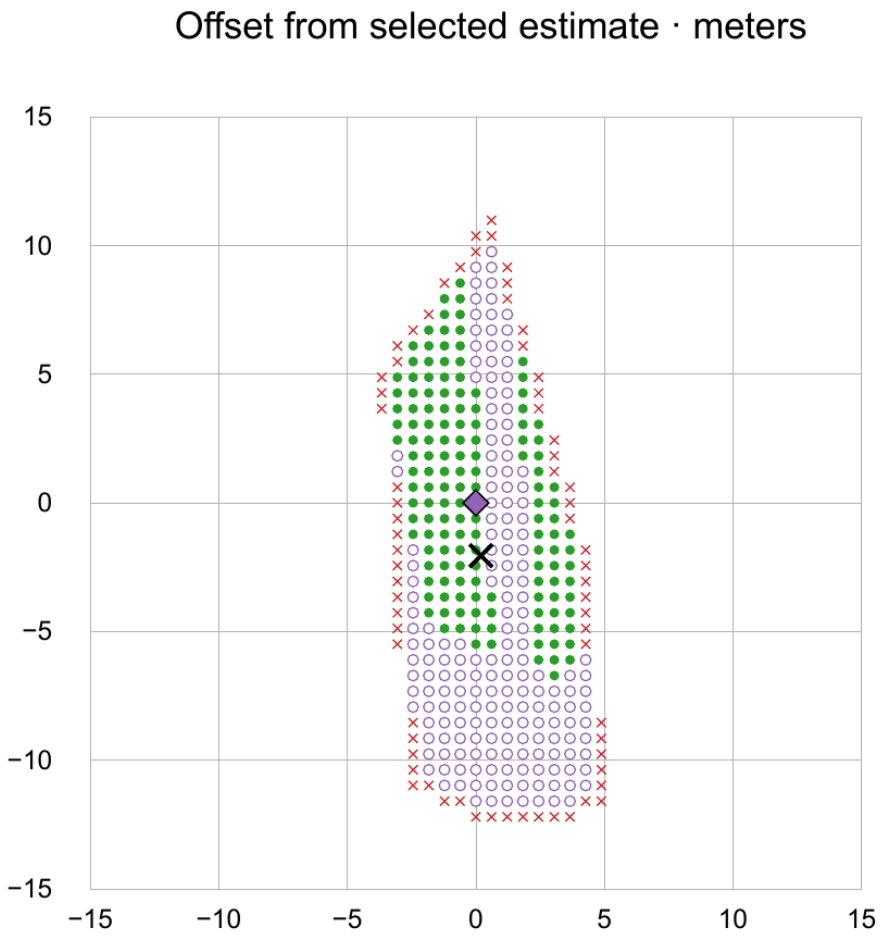
2.06 m

64 / 64 aid bearings matched

× Truth ◆ Selected estimate

Simulated observations on archived NOAA geometry.

The remaining local alternatives



148

Consistent centers

67

Inconsistent centers

182

Unresolved centers

× Truth ◆ Selected estimate

Chart aids versus chart aids plus AIS

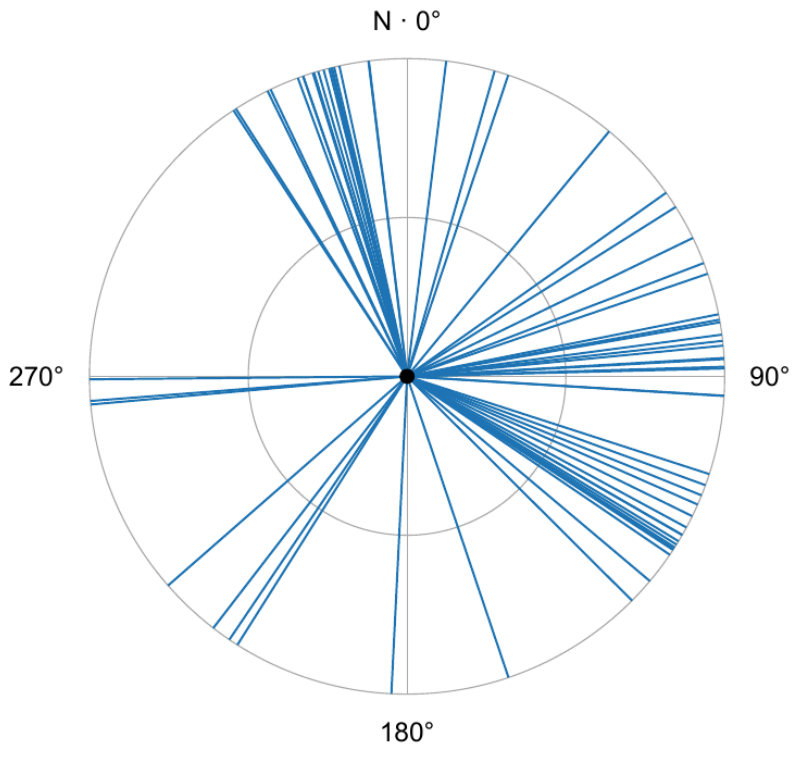


Chart aids

Proposed comparison

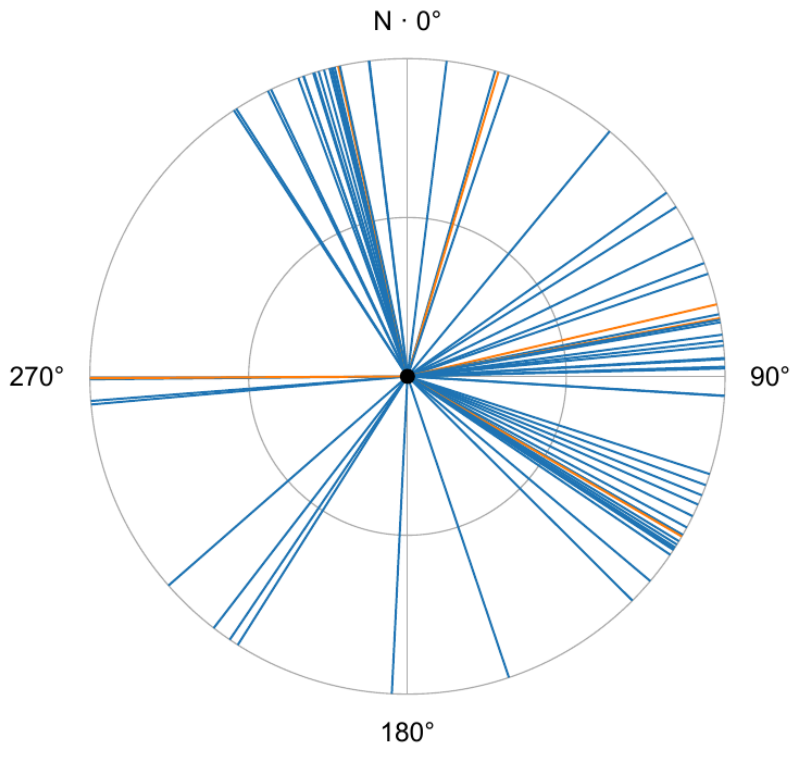


Chart aids + AIS

Proposed evaluation · current result uses chart-aid bearings.

Time to locate the observer



Saved simulated Charleston frame · independent truth evaluation

8.46 s

Search wall time

Eastern Seaboard

3 GPUs

Simulated observations on archived NOAA geometry.

Current capability and next evaluation



Saved simulated Charleston frame · independent truth evaluation

2.06 m

Position error

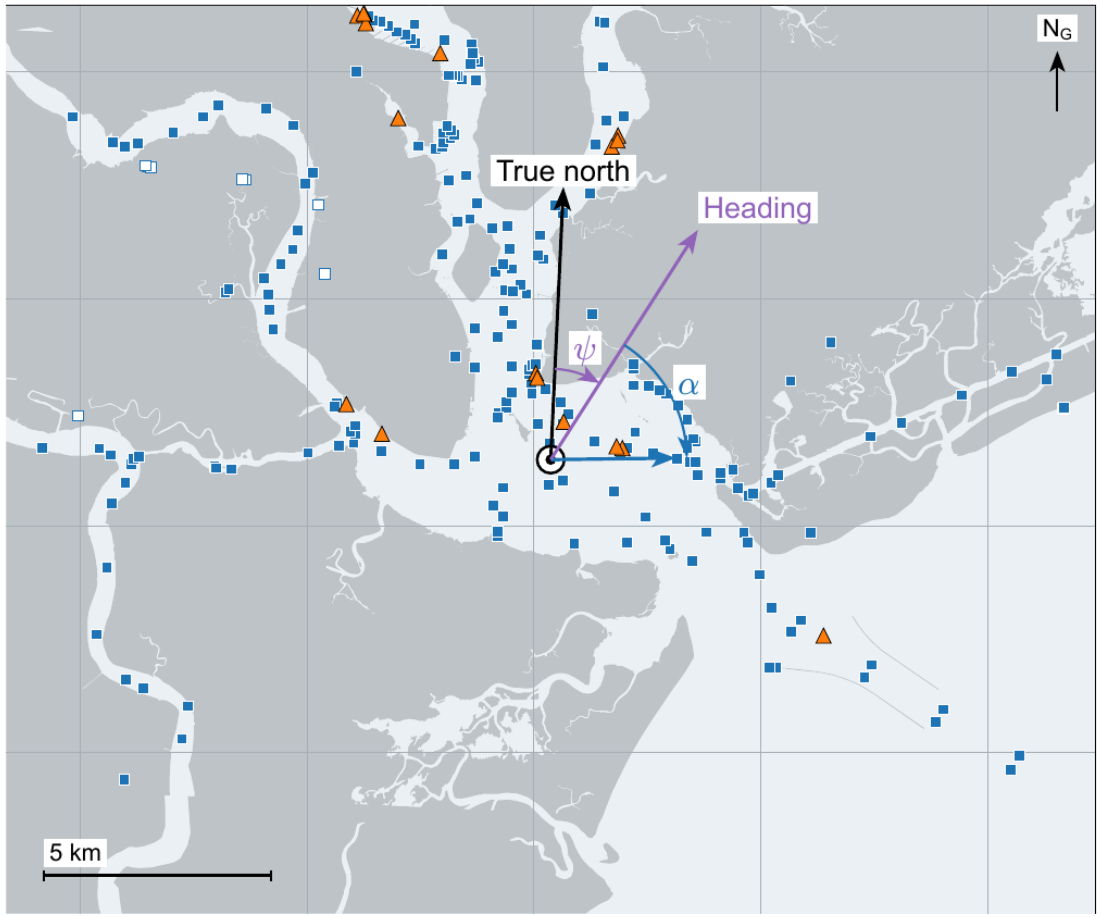
8.46 s

Full-coast search

64 chart-aid bearings

Simulated observations on archived NOAA geometry.

Bearing and coordinate conventions



$$\beta = \mathrm{wrap}(\psi + \alpha)$$

ψ · known heading

α · relative bearing

β · global bearing

Heading-conversion example · calculated $\psi = 30°$

Complete sensor specification

Coverage / radius	360° / 10 km
Angular grouping	0.1°
Gaussian σ / hard bound	1/6° / ±0.5°
Detection probability	0.9
Type errors / bearing outliers	0 / 0
Extra vessels	Poisson mean 1 · realized 1

Whole-cell uncertainty envelope

$$E_{\text{true}}(C) \subseteq E_{\text{possible}}(C)$$

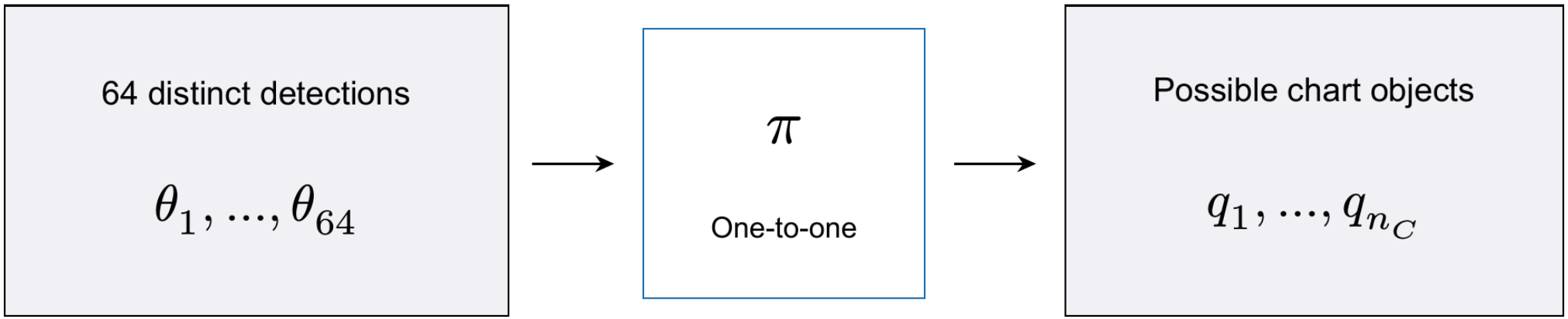
Possible edges cover every physical explanation within the cell.



$$\nu\big(E_{\text{possible}}(C)\big) < 64 \Rightarrow C \ \text{ is rejected}$$

Unknown bounds, identity or capacity → retain.

Injective feasibility proof



$$i \neq k \Rightarrow \pi(i) \neq \pi(k)$$

$$|N_C(S)| < |S| \Rightarrow \nu(E_C) < 64$$

Equations only. Actual rejection adjacency awaits a recorded witness.

Exhaustive count arithmetic

Calculated work counts

$$1644677 \times 2388460 = 3928245227420$$

1 m rectangular grid positions

$$39435 \times 4^{14} = 10585752207360$$

Terminal cells if every root is refined without pruning

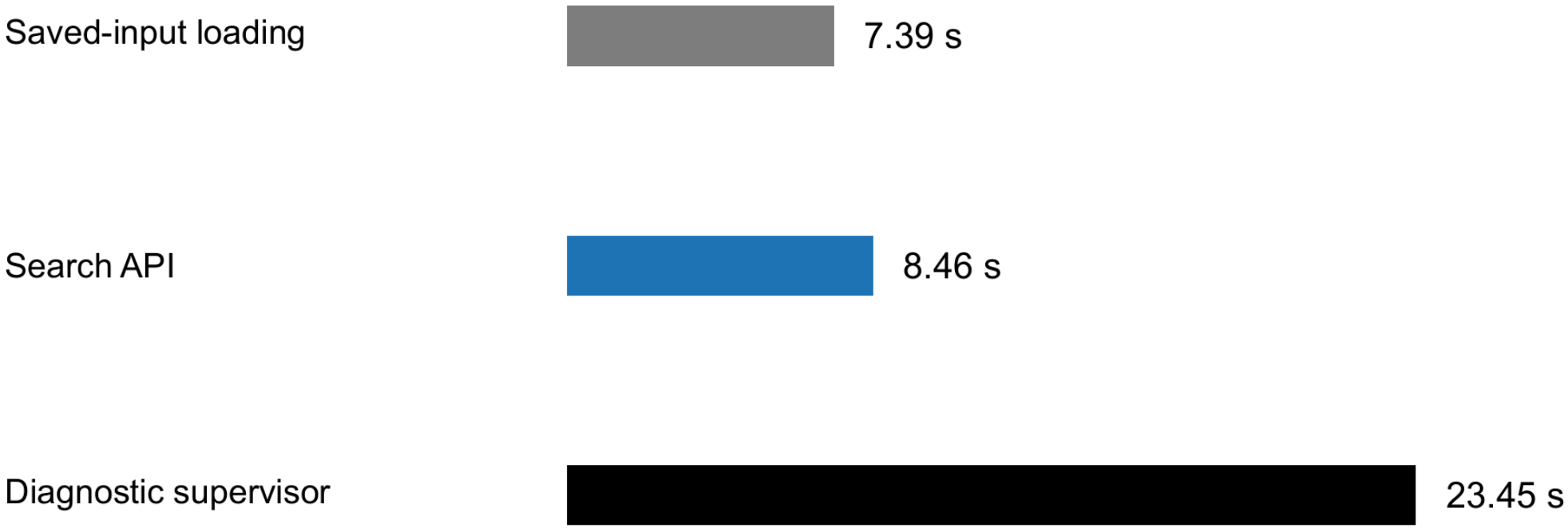
14,114,336,263,335 total tree nodes · no measured exhaustive runtime

Catalog provenance and epochs

15 July 2023	AIS snapshot · 12:00–12:01 UTC
24 October 2026	ENC chart-point retrieval
26 October 2026	Saved shoreline export

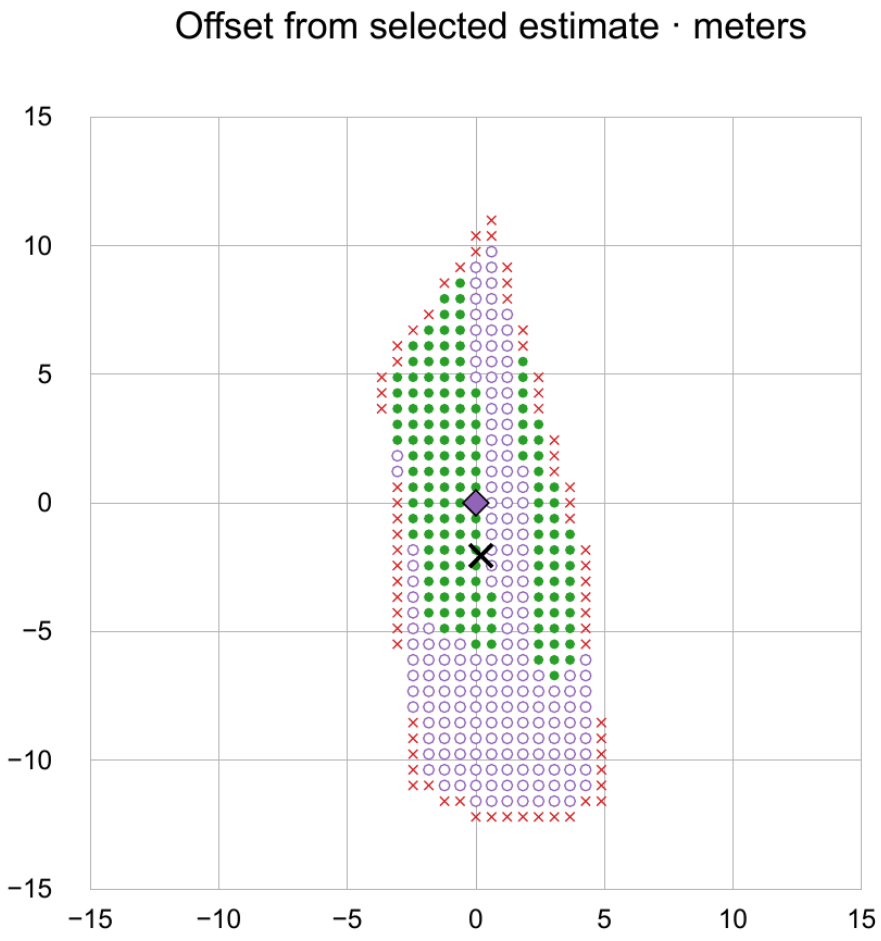
Mixed epochs · archived geographic test world

Timing definitions



Different scopes · saved cache was read, not rebuilt

Remaining unresolved centers



148

Consistent centers

67

Inconsistent centers

182

Unresolved centers

× Truth ◆ Selected estimate